higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS TECHNOLOGY
NQF LEVEL 2

27 FEBRUARY 2019

This marking guideline consists of 5 pages.

Copyright reserved

QUESTION 1

1.1 False – wall thickness
1.2 False – rotary pump
1.3 False – final products
1.4 False – surface/depth filter
1.5 False – adsorbate

(5 × 2) **[10]**

QUESTION 2

2.1
- Process piping
- Major equipment
- Control valves and valves that affect process operation
- How systems are interconnected
- Major bypass
- Reticulation streams
- Operational data (like temperature, pressure, mass flow rate)
- Process stream names (Any 5 × 1) (5)

2.2 Indicate flow through pipes (2)

2.3 By various symbols (2)

2.4
- Main process
- Minor process
- Insulated pipe
- Jacketed pipe
- Pipe with heat tracing
- Electrically heated pipe (Any 1 × 1) (1)

2.5 To transport fluids between equipment in a process plant (2)

2.6
- Steel
- Stainless steel
- Copper
- Aluminium
- Polymers (5)

2.7 Union (1)

[18]

Copyright reserved

QUESTION 3

3.1 3.1.1 When liquids and solids are separated using gravity
OR
Removal of suspended solids from effluent water due to gravity

3.1.2 Lighter, smaller suspended solids that float on top of a tank

3.1.3 Process where microorganisms are destroyed inside water
OR
Process of killing only harmful pathogens inside water

(3 × 2) (6)

3.2
- Carbon dioxide (CO_2)
- Methane (CH_4) (2)

3.3 Decks (2)

3.4
- Ultrafiltration
- Smoke generator
- Steam generator
- Water purifier
- Lower-pressure applications and general- or oil-service applications
- Continuous-duty piston pumps in power applications where the piston is in contact with water
- Indexing, conveying, lifting, clamping and machine-feeding or extracting applications (Any 4 × 1) (4)

3.5
A Discharge line
B Impeller
C Diffuser plates
D Volute
E Suction line
F Suction eye

(6 × 1) (6)

3.6 3.6.1 Used as a space within which a piston travels
3.6.2 Used to exert a force on the liquid inside the cylinder

(2 × 2) (4)

3.7
- To fill finished product packages accurately
- To determine receipts
- To dispatch quantities of various types of materials
- To measure the correct quantity of materials making up a batch material used in the process (4)

[28]

QUESTION 4

Copyright reserved

4.1
- Always read the label of the item to be stored to determine if there are any specific requirements.
- Separate items that can react with each other.
- Items stored in boxes or porous containers should not be exposed to wetness or condensation as this will ruin the products.
- Some products must be stored upright or on their sides.
- Some items must be stored in the dark as exposure to sunlight can damage them.
- Some items must be stored at room temperature and should not be exposed to heat or extreme cold.
- Old material should not be used and must be disposed of if expired.
- Ensure that items with caps or leads are tightly closed to avoid exposure to air which may ruin the product. (8)

4.2 4.2.1 Plastic or stainless steel (2)

4.2.2 Storage and transportation of quantities of materials from the producer to the client (2)

4.3 Because they are highly explosive, corrosive or poisonous (2)

4.4
- Protects against corrosion
- Protects against erosion
- Prevents the growth of microorganisms
- Improves the appearance of the equipment (4)

[18]

QUESTION 5

5.1 5.1.1 Process that removes suspended solids and particulate matter from a liquid

5.1.2 Industrial decantation process that separates substances by continuously refeeding partially decanted products back into the system

5.1.3 Process using centrifugal force to separate mixtures of different densities

5.1.4 Process using both gravity and centrifugal force to separate solid particles from a gas or liquid stream

5.1.5 Process of producing crystals from a solution

(5 × 2) (10)

5.2
- Purification of gold and other metals
- Noble gases
- Medicine preparation
- Removal of bacteria and viruses from sewage
- Air filters in gas masks and respirators (Any 4 × 1) (4)

5.3 The longer the fluid remains in the tank,✓ the better the sedimentation process will be.✓ The higher the flow rate,✓ the bigger the tank should be to give sufficient retention time for the solids to settle.✓ (4)

5.4
- Water treatment
- Separate milk from cream and oils from water (2)

5.5 5.5.1 Scrapes the layer of solid build-up from the basket while the basket continues to spin

5.5.2 Sprays the wash liquid onto the solids to wash out the liquid trapped in the solids

(2 × 2) (4)

5.6 To protect downstream equipment from contraries such as stones and metal fragments (2)

[26]

TOTAL: 100

Copyright reserved